

The Role of Subject-Matter Knowledge and Interest in the Processing of Linear and Nonlinear Texts

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Sixty-six studies were reviewed that met several a priori criteria. Specifically, the studies had to be empirical investigations that related to a particular academic domain and that involved connected discourse presented either in traditional written form or on computer. In addition, the studies had to incorporate some measure of both knowledge and interest. The resulting body of literature was first summarized and analyzed in terms of the domains chosen, the subjects selected, the nature of the texts used, the manner in which knowledge and interest were assessed, and the principal outcomes reported. Next, from this analysis, six premises were proposed as guides for future research and practice. Finally, concluding remarks were advanced that address the overall significance of text-processing research that interactively considers the domain of knowledge and the interest of the reader.

Recent research has brought increasing attention to the importance of subject-matter knowledge in human learning and development (e.g., Chi, Glaser, & Farr, 1988; Glaser, 1984; Glaser, Lesgold, & Lajoie, 1987). As this research has indicated, it is difficult to overestimate how much individuals' prior knowledge contributes to future learning within a particular field of study. Yet, even more recently, the role of subject-matter knowledge has been juxtaposed with other cognitive and affective variables for the purpose of examining interactive effects (Alexander, 1992; Dochy, 1992; Perkins & Salomon, 1989; Pintrich, 1989; Prawat, 1989). For those engaged in this interactive, domain-related research, it is not a question of whether existing subject-matter knowledge influences subsequent learning. Such an influence is assumed. Rather, for these researchers, it is a question of how additional factors in the learner or the learning environment (e.g., personal interest or instructional materials) mediate or mitigate the effects of that knowledge.

Two examples may help to illustrate what we mean by interactive, domain-related research. Alexander and Judy (1988) synthesized literature on the inter-

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action of domain knowledge and general strategy knowledge in academic performance. Their purpose was to understand better how these two powerful forces might operate in consonance or dissonance. One of their findings was that strategic processing was more effective and efficient when individuals possessed at least a foundation of domain knowledge. They also posited that the nature of strategy use changed as individuals became more knowledgeable in a domain.

In another interactive investigation, Schiefele, Krapp, and Winteler (1992) studied the effects of interest on academic achievement by means of a meta-analysis. Schiefele et al.'s aim was to identify patterns across a body of research that might clarify how motivational factors impact academic learning. One of their conclusions was that gender seems to play a role in this relationship. What they uncovered in the studies they reviewed was a significant difference between males and females with regard to the relationship between interest and academic achievement. Specifically, interest explained 12% of the reported variance for males but only 6% for females. Again, these are just two illustrations of the research that has considered subject-matter knowledge in association with other influential factors in the learner or the learning environment.

It is the goal of this review to expand on the aforementioned research and other studies by examining literature that investigates subject-matter knowledge interactively. That is, we will consider subject-matter knowledge in relation to the form of text used in instruction and in conjunction with learner interest. From an educational standpoint, we believe that such a review is significant because it allows one to understand better the way that one's knowledge in a particular field interplays with variables that have been judged as critical to learning, text, and interest. Likewise, from a theoretical standpoint, it is our hope that the premises we derive from this review can serve as guides for future multidimensional studies of subject-matter knowledge, text, and interest. In many ways, we feel, as do others (e.g., Pintrich & Schrauben, 1992; Salomon, Perkins, & Globerson, 1991), that educational research has matured beyond the stage where theory and practice are well informed by studies that investigate single variables in controlled or contrived ways.

Specifically, this review of the literature focused on two general types of texts, linear and nonlinear. We centered our review on text for several reasons. First, we have been actively engaged in text research for a number of years and are familiar with literature in that realm. Further, we found that many researchers exploring subject-matter knowledge from a complex, interactive perspective used text as a central component in their investigations (e.g., Hidi & Baird, 1986; Hidi, Baird, & Hildyard, 1982; Salomon, Globerson, & Guterman, 1989; Tyler & Voss, 1982; Walker, 1987). Finally, from the field of science (Yore, 1991) to the domain of social studies (Beck & McKeown, 1988), text remains a principal means by which individuals acquire subject-matter knowledge (Chall & Conard, 1991).

We use the term *linear text* to designate connected discourse presented in written form where decisions relative to processing are left solely to the reader. Defined in this manner, linear text is text of a more traditional nature—that is, the writings students are apt to encounter in textbooks, journals, and magazines. *Nonlinear text* is also connected discourse, however; it is discourse accompanied

by some type of database management system. This system guides or prompts readers to reaccess or extend the main text through associative computer-based links to other informational screens (see Gillingham, Young, & Kulikowich, in press, for an extensive discussion of the nature of nonlinear text). One category of nonlinear texts that we will inspect in this review, for example, is hypertext. With the evolution of situated theories of learning that emphasize rich and unique learning environments (e.g., Brown, Collins, & Duguid, 1989; Greeno, 1989; Resnick, Levine, & Teasley, 1991), nonlinear texts, such as hypertext, have gained in popularity. Many of these hypertexts are part of extensive multimedia systems that include not only nonlinear texts but also supporting videos, maps, and commercial movie clips (Christensen, Giamo, & Jones, 1993; Trumbull, Gay, Mazur, 1992). Therefore, it is important to consider the impact of these more nontraditional texts on the acquisition of subject-matter knowledge (Bolter, 1991b).

As with others (e.g., Pintrich & Schrauben, 1992; Weinstein & McCombs, in press), we acknowledge the influence of noncognitive variables in one's development within an academic domain. In deciding how to incorporate the noncognitive (i.e., motivational) dimension to this review of subject-matter knowledge, we decided to concentrate our efforts on the construct of interest. This decision was made, in part, because of the inconsistent findings regarding the relationship between interest and knowledge that have been reported in the literature (Schiefele et al., 1992; Tobias, in press). In addition, our own research has recently led us to question the role that interest plays within various stages of domain learning (Alexander, in press; Alexander, Jetton, & Kulikowich, 1993). For the purpose of this review, we defined *interest* to be both situational and individual (Hidi, 1990; Schiefele, 1991). By situational interest, we mean the transient attention or emotional response to elements within the text or the context. Individual interest, on the other hand, refers to a more long-term, deep-seated interest in the domain from which that content arises.

In the sections that ensue, we first articulate the criteria and the procedures we followed in conducting this review. Next, we summarize and analyze the literature we identified, and then we forward basic premises about the interrelationship of subject-matter knowledge, interest, and linear and nonlinear text that we hope can guide future endeavors. Finally, we offer concluding remarks that speak to the significance of subject-matter research that attends interactively to learner interest and to the nature of the instructional text used in the learning environment.

Method

Criteria for Inclusion

Our first task in this review was to specify the criteria that would serve as our decision rules for the inclusion or exclusion of particular literature selections and to stipulate the procedure to be followed in locating and summarizing each work. Following extended discussions, we determined that literature would be included in this review if it met four criteria. First, whether quantitative or qualitative in approach, the work had to be empirical. Reviews of the literature, like Hidi's (1990) synthesis on interest, while useful as a source of relevant research, leading

questions, or guiding hypotheses, were not explicitly analyzed in this review. Second, there had to be an academic domain of study that was either specified by the researchers or that we could discern from the given information, such as the descriptions of the experimental materials. Studies, such as Bromley and Mahlios's (1985) investigation of good and poor readers' text-processing strategies, were not included because they did not offer adequate information on the domain from which the text was drawn. Such a problem was rather common in studies in the fields of reading and writing and led to the exclusion of a number of investigations in our initial pool.

Third, text had to be a component in the study, and, further, that text had to be connected discourse. Thus, studies that centered on isolated words or sentences were excluded from our review. This criteria resulted in the elimination of only a small number of studies from our initial pool, among them Anderson's (1982) work on selective attention for which the experimental stimulus was a list of unconnected sentences. Although the study had to involve discourse, the text could be presented in either linear or nonlinear form. The comparison of these two forms of text was an important dimension of this review because these forms of discourse were indicative of more and less traditional learning environments. Some recent work by the Cognition and Technology Group at Vanderbilt (1990, 1992) on videodisc environments, while relevant to learning environments, was not summarized because it did not specifically incorporate text as we have defined it in this review.

Fourth, both knowledge and interest variables had to be studied. We frequently found literature that addressed either knowledge or interest but relatively little that specifically examined both. In a study by Shirey and Reynolds (1988), for example, there was an investigation of the effects of interest on text learning. However, there was no measure of subject-matter knowledge involved in the study, excluding it from our analysis. Reciprocally, Walker's (1987) excellent investigation on the relative importance of domain-specific knowledge and general aptitude to subject-matter learning could not be analyzed, because it did not include any measure of interest. There were also several cases, particularly in the domains of mathematics and science, where the researchers investigated participants' knowledge and attitudes but did not include any specific measures, subscales, or items that related directly to interest. Henk and Holmes's (1988) research on the effects of content-related attitudes on comprehension and Schibeci and Riley's (1986) examination of the influence of background and perceptions on students' attitudes and achievement in science were studies eliminated from the review for this reason. Overall, this particular criterion, more than any of the preceding three, resulted in the exclusion of studies in our initial pool from final review.

Search Procedure

A general procedure was followed in locating relevant literature. First, we generally restricted our search to the period extending from 1983 to the present because we felt that most pertinent literature would have been reported within the past decade. However, we decided to include any research reported prior to that date if it were frequently cited or if we knew it to be central to our review. Hidi et al.'s (1982) classic study, entitled "That's Important But Is It Interesting?"

Two Factors in Text Processing,” is an example of a study published before 1983 that we incorporated in our review. Next, we examined a number of on-line retrieval systems, including Psychological Abstracts, Educational Resources Information Center, and Dissertation Abstracts. Key words that were used to identify prospective writings included the term *knowledge*, or such knowledge-related descriptors as comprehension, recall, achievement, prior knowledge, or learning, and the term *interest*, or such interest-related descriptors as attitude, affect, intrinsic motivation, and preference. Even though we used related terms, such as attitudes, to search for relevant literature, we included only those writings that fit the operational definitions of subject-matter knowledge and interest we offered at the outset of this review.

Along with these database retrieval systems, we physically scanned the volumes of 65 journals from various fields. The list of journals examined are listed in Appendix A. Again using key words, we perused the title and abstract of each article. If additional information were required to determine the suitability of an article, then the materials section and the description of the variables were also inspected. Finally, using the same process, we examined the contents of books, technical reports, and papers from conferences, such as the American Educational Research Association. On the basis of this search procedure, we identified an initial pool of over 750 published and unpublished works that seemed to include some combination of variables of concern in this study (i.e., knowledge, text, and interest). From this initial pool, 66 studies met the criteria for inclusion we specified earlier. These studies are summarized and discussed in the section that follows.

Summary and Critical Analysis

To assist us in analyzing the 66 studies identified for this review, we first summarized them with regard to central characteristics. Specifically, as seen in Appendix B, we recorded:

- the authors and the year reported,
- the domain or domains researched,
- the number and type of subjects,
- the nature of the text (i.e., whether linear or nonlinear),
- a further description of that text (e.g., length and topic),
- when knowledge was measured (i.e., pretest or posttest),
- how knowledge was assessed,
- whether interest was pretested or posttested,
- the type of interest measure used, and
- the principal outcomes of the research.

In examining these basic characteristics, we identified some general trends across the studies. For example, over 75% of the studies meeting our stated criteria have been reported in the last 5 years. This is especially true for the research involving nonlinear text, the vast majority of which has appeared since 1990. In light of the increasing sophistication in domain-related research and the recency of multimedia approaches, such as hypertext, this trend is quite understandable. Additional trends are summarized and analyzed under their respective headings.

Domains

Despite the number of domain-related journals we examined, we found that those researchers exploring subject-matter knowledge, in conjunction with text processing, tended to cluster in a relatively few fields. Predominantly, identified studies involved texts drawn from the domains of literature, psychology, social studies, and science. Relevant studies were also located in the domains of mathematics and computer science. It might be argued that this clustering of domains is partly the result of our decision to restrict this review to research involving connected discourse. While this argument cannot be discounted, it is not a sufficient explanation for this overall trend. It is likely that several other conditions contributed to this pattern.

For one, literature may have been a preferred domain because it is the primary medium by which students learn to read in schools (Purvis, 1981), thus making it particularly appealing to researchers in language and literacy. For another, it seemed that literature was a domain selected by researchers working with both linear and nonlinear text because of its ill-structured nature (Bolter, 1991a). This ill-structured quality not only allowed for the study of cognitive processing or strategic behavior (e.g., Salomon et al., 1989) but also for what Beeman et al. (1987) labeled *pluralistic thinking*. *Pluralistic thinking* is a term that Beeman et al. used to designate the critical analysis of concepts in one domain from the perspective of another domain. For instance, one can analyze the writings of Dickens from the perspective of Victorian literary style or from the social class issues that were operating at that time.

A similar interdisciplinary explanation could also be applied to some of the research we reviewed in social studies. By using passages drawn from social studies, however, researchers (e.g., Hidi & Baird, 1988; Wade & Adams, 1990) were not only able to stimulate pluralistic thinking, but they were also able to include an array of topics (e.g., “Easter Island” and “Inventors”) likely to interest readers. In addition, those working with social studies texts were able to manipulate certain text variables, such as importance and interestingness of specific information, as well as the overall structure of the text (e.g., narrative, pseudonarratives, and exposition). These types of manipulations were central to the researchers’ questions of how the nature of the text influenced subjects’ learning.

The frequency of relevant studies from the domain of science appeared to be linked with the current interest in students’ misconceptions (e.g., Morrell, 1992). Some researchers used what Carlsen and Andre (1992) described as *conceptual change* texts. These texts, which typically included both linear and nonlinear components, were intended to confront common misconceptions in such areas as genetics or electric circuitry. Thus, the text was an important tool for conveying accurate and detailed information about complex concepts that were difficult for students to comprehend accurately.

Another domain where our studies tended to cluster was mathematics. In several of these mathematics studies where computer-based or computer-assisted instruction was utilized, the text was in the form of extended word or story problems. Here, the choice of domain seemed more related to students’ interests in the field than to their knowledge base or the nature of text processing.

Specifically, what researchers like Pokay and Blumenfeld (1990) were most interested in was why students found mathematics boring or unengaging. One exception to this observation was the program of research by Ross and colleagues (e.g., Murphy & Ross, 1991; Ross, McCormick, & Krisak, 1986). In this research, Ross and his colleagues were quite concerned with the nature of the text because they wanted its contents to be closely aligned with students' interests. They believed this alignment would stimulate academic performance. For example, Murphy and Ross (1991) sought to make mathematics more meaningful to students by constructing story problems with themes of personal interest to the students (e.g., animals, sports, and clothing).

Finally, pertaining to the studies in psychology, we have two observations. First, it appeared as if the choice of psychology was most often a pragmatic decision rather than a theoretically driven one. That is, in these investigations, the researchers may have chosen a content-area of convenience because no explicit or implied explanation was offered as to why this content was considered over any other (Benware & Deci, 1984; Schiefele, 1991, Exp. I). It might be surmised, however, that, once the decision was made to use college students, the choice of psychology texts was reasonable because the students could read texts that were authentic (i.e., actual chapters in textbooks) or relevant to their college studies. Of course, this was not always the case. Students were occasionally asked to read and react to written materials that were both highly contrived and rather irrelevant to their academic pursuits (e.g., Garner, Alexander, Gillingham, Kulikowich, & Brown, 1991), although this observation applies more to studies using linear than nonlinear text.

Subjects

Studies employing more traditional text covered the spectrum of school-age subjects, whereas those requiring subjects to manipulate the text via a database system most often studied students of high-school and college age. When database systems were used with younger students (i.e., elementary or middle-school students), these systems were often utilized to promote written composition processes (e.g., Turner & Dipinto, 1992) or to foster cooperative learning strategies (e.g., Signer, 1992). It is likely that the demands of navigating through the computer-based system, particularly hypertext, is too challenging for younger learners, restricting the applicability of such programs. Indeed, in one study (Egan, Remde, Landauer, Lochbaum, & Gomez, 1989), the researchers hired experts in statistics so that they could study these experts' interactions with a hypertext system and the search strategies which they employed. (As above, we will make reference to students' navigation through nonlinear text. By navigation, we are referring to the readers' movement from one segment of text to another or from the text proper to other screens, tools, or options, such as supporting videos, maps, and commercial movie clips.)

Further, in relation to the subjects, we were somewhat surprised by the sample sizes. Perhaps because they tended to use computer classes as their source of subjects or because of the ease of data collection, approximately 60% of the nonlinear text studies employed large samples (range: 20–252). Even more of the linear text studies had large samples (74%), and their range in sample size was comparable (range: 20–540).

Beyond this basic information, relatively few researchers offered rich descriptions of the individuals they studied. Occasionally, gender or ethnic background was referenced (e.g., Anderson, Wilkinson, & Mason, 1991; Kinzie, Sullivan, & Berdel, 1992), but even this information was unavailable for a number of the investigations, particularly those involving older subjects or those summarized in edited volumes (e.g., Wade, 1992). When we consider that gender has been suggested as a significant factor in the interplay of knowledge and interest (Schiefele et al., 1992), the failure of researchers to consider such information seems problematic. For our purposes, the lack of data was particularly acute in terms of the background knowledge and experiences of the subjects. As we will discuss later, researchers often made assumptions about subjects' relevant knowledge, based on age or grade level, but failed to provide empirical support for those assumptions.

Texts

It is interesting to note that the linear and nonlinear texts described in these studies varied on a number of attributes. For instance, easily over one third of the traditional texts were expressly written, manipulated, or contrived for the particular research study; a trend not evidenced with nonlinear texts. Garner, Gillingham, and White (1989), for example, structured brief exposition on insects to highlight particular "seductive details" (p. 41). Likewise, Garner et al. (1991) manipulated a passage on Stephen Hawking to strategically position personally involving content dealing with his battle with a life-threatening illness.

With nonlinear text, there was a greater tendency for the researchers to integrate a variety of original, resource texts (e.g., journal articles and scientific papers) with visual and graphic aids. With some exceptions (Anand & Ross, 1987; Morrison, Ross, & Baldwin, 1992), there was much less manipulation or generation of text in these studies, unless that manipulation or generation was done by the subjects, who were free to create a text using a collection of source materials (e.g., Egan et al., 1989). In addition, the texts used in the computer-based or hypertext investigations were characteristically lengthy—much more so than was typical in the studies using more traditional texts. The length of these nonlinear texts may be another reason why subjects in these studies tended to be older.

We were intrigued by the number of researchers who used modified or unmodified excerpts from encyclopedias in their investigations (e.g., Asher, Hymel, & Wigfield, 1978; Baldwin, Peleg-Bruckner, & McClintock, 1985; Wade & Adams, 1990). Perhaps their decisions to use passages from encyclopedias were based on the accessibility of these reference materials in schools and, thus, students' familiarity with them. Indeed, encyclopedias are ready sources of short exposition on a wide array of topics. Yet, this frequent use of texts from encyclopedias also concerned us for several reasons. Despite their popularity, we are not convinced that entries in encyclopedias are particularly well crafted as text, nor are we sure that these reference materials present a topic in any depth or detail. Therefore, students asked to process passages drawn from encyclopedias may be required to learn from incoherent or unengaging texts that treat a topic in a rather superficial manner.

Regrettably, in a number of studies, little information beyond a cursory mention of the text was provided (e.g., Dillon, 1991). With these investigations,

we were left with the impression that the structure or content of the text was viewed as a nonsignificant factor in the outcome of the research. However, in those studies where the researchers purposely wrote or manipulated the text, much more information was typically given as to why that particular topic or passage was chosen or why certain features of the text were judged important. This may have to do with the fact that these researchers, many from the fields of reading or educational psychology, were much more concerned with general text processing than with domain-specific issues. For these individuals, the selection of a domain-related passage was, in essence, a secondary or tertiary issue. The richer text descriptions among these researchers may also be related to the fact that some chose texts to correspond to subjects' reported likings or preferences (e.g., Hidi & Baird, 1988). Therefore, more elaborate descriptions of the texts were required to substantiate this correspondence.

Knowledge. Whether the text was linear or nonlinear in nature, knowledge was in many cases assessed in a post hoc rather than in an a priori fashion (e.g., Wade, 1992). For the most part, researchers seemed more interested in measuring subjects' comprehension or recall following the processing of text than they were in gauging subjects' preexisting knowledge relative to the selected field or text. As we noted, some researchers (including these authors) assumed that students, as a consequence of their age or their educational level, would possess the general or domain-specific knowledge necessary to process the given content effectively (Alexander, Jetton, Kulikowich, & Woehler, in press; Jetton, Alexander, & White, 1992; Morrell, 1992). In retrospect, such assumptions seem risky. For example, in a recent study, Alexander, Kulikowich, and Schulze (in press-a) found that undergraduates reported almost no knowledge about the standard model in physics and relatively little knowledge about Stephen Hawking and grand unification. While the researchers expected the reading tasks to be somewhat demanding, they never anticipated the dearth of physics knowledge they encountered among educated students.

In the studies that deviated from this general pattern (e.g., Alexander, Kulikowich, & Schulze, in press-a, in press-b; Faircloth, 1984; Schiefele, 1992b, Exp. I; Steele, 1985; Walton, 1982), pretest measures were occasionally administered either to control for or to categorize subjects on the basis of their knowledge. In the case of Schiefele's (1992) experiment involving computer-science majors' processing of psychology texts, the goal was to ensure that subjects had little preexisting knowledge of the textual content. By contrast, Alexander, Kulikowich, and Schulze (in press-a) categorized students as *low knowledge* or *high knowledge* on the basis of their pretest performance on a domain knowledge test. This knowledge differential proved to be a significant factor in both the students' interest in the content and their recall of the passage information. Prior subject-matter knowledge also proved to be a significant variable in the study by Garner et al. (1991). In this study, students' level of domain knowledge militated against the personally involving information included in a passage about Stephen Hawking and the grand unification theory. Further, recent work by Scardamalia and Bereiter (1992) showed that the familiarity of students with the topic of endangered species helped those who did not have text materials on the topic ask more knowledgeable and more interesting questions than students who asked questions on the topic after reading a related text. Schumm, Mangrum, Gordon,

and Doucette (1992) also found that students with high knowledge and interest were better able to predict the questions that their professors would ask.

Clearly, a variety of knowledge measures were used, including free recalls, grade-point averages, achievement test scores, short-answer and open-ended questions. However, the most common format for assessing students' knowledge, including both standardized and researcher-generated measures, was via multiple-choice questions. It is likely that ease of scoring was one factor that contributed to the popularity of this measurement format, along with the usefulness of multiple-choice items in covering a breadth of subject matter. Further, while most of the measures assessed declarative knowledge, there were some procedural knowledge measures applied, particularly in the computer-based studies (e.g., an assembling task; Dillon, 1991; a figural construction task; Morrison et al., 1992). We observed that there was an apparent relationship between the domain under study and the preferred knowledge measure, with researchers in science and mathematics often choosing standardized achievement data in contrast to those in reading and literature who chose structured and unstructured recall measures.

Interest

Whereas knowledge was most often measured post hoc, interest was often assessed both at the outset of the research and at its conclusion. When measures of interest were taken prior to text processing, they were often done to determine students' preferences for particular topics (e.g. hobos; Hidi et al., 1982) or for a subject-matter area (e.g., meteorology; Gardner, Simmons, & Simpson, 1992). Such measures of interest are more aligned to what Hidi (1990) labeled individual interest. By comparison, when researchers had subjects report their interest during or immediately following text processing, they were frequently concerned with aspects within the immediate situation that produced temporary excitement, attention, or interest (Garner & Gillingham, 1992; Sadoski, Goetz, & Fritz, 1993). This has appropriately been referred to as situational interest. Those measurements that were a bit more difficult to classify as either individual or situational interest were those that were associated with interest scales administered as part of mathematics, science, or computer studies, or with attitude measures containing interest items (e.g., Jegede, Okebukola, & Ajewole, 1991; Morrell, 1992). In these cases, such items (e.g., "Computer Assisted Instruction [CAI] makes learning interesting") appeared to incorporate aspects of both situational and individual interest, depending on the respondents' interpretation of intent; that is, did the student respond in relation to the task at hand (e.g., the specific computer program) or in terms of a long-term orientation to the content (e.g., computers or computer-based learning in general).

As was the case for knowledge, the variety of measures used to assess interest was extensive, ranging from sentence underlining (e.g., Alexander, Kulikowich, & Schulze, in press-a), to ratings of interestingness (Wade & Adams, 1990), to questions inquiring about subject-matter interests or topic preferences (e.g., Murphy & Ross, 1991). Whether ratings, surveys, or questionnaires were used to assess interests or preferences, many researchers (e.g., Monroe, 1991) used a Likert-type scale that ranged from 1 to 5 or 1 to 10 points. As we mentioned previously, with these Likert scales, the variable of interest was sometimes

restricted to just one or two items from a larger attitudinal measure (e.g., Gray, 1989; Lanza & Roselli, 1991; Morrell, 1992). This approach, which occurred most often in the computer-based studies, resulted in very limited information on subjects' interest in the domain or the experimental task.

As we noted, the assessment of interest was sometimes made at a global level in relation to a field of study (e.g., science; Romance & Vitale, 1992), a particular topic (e.g., inventors; Hidi & Baird, 1988), or the use of technology (e.g., Gardner et al., 1992; Guthrie & McPherson, 1992). At other times, however, the researchers were concerned with gauging subjects' immediate interest in a particular text selection (e.g., quarks; Alexander, Kulikowich, & Schulze, in press-b) or some segment of that text (e.g., wager between Hawking and Kip Thorne; Garner et al., 1991). It is noteworthy that in several studies where traditional texts were used, particularly the seductive detail research (e.g., Garner et al., 1989; Wade & Adams, 1990), subjects' recall was used as an indicator of interest. In the Garner et al. (1989) study, for instance, students' recall of specific information from text was contrasted against experts' ratings of interest and importance for those same text segments. The students never explicitly rated or discussed their interests.

Outcomes

Five trends with regard to the principal outcomes reported in our 66 studies are especially noteworthy. First, although we made the a priori decision to include both quantitative and qualitative studies, our literature review was composed almost exclusively of quantitative research (i.e., 65 out of 66 studies). It may well be that the research question we posed regarding the role of subject-matter knowledge and interest in the processing of linear and nonlinear texts is a question most likely considered or analyzed quantitatively. It might also be that, as quantitative researchers ourselves, we have missed rich qualitative sources that speak to this empirical question.

Second, and related to the previous issue, researchers overwhelmingly employed group comparison procedures such as analysis of variance and analysis of covariance to determine their outcomes. Within-subject perspectives among these studies were, with rare exceptions (Alexander et al., 1993; Anderson-Inman & Horney, 1993), forgotten or ignored. As Alexander et al. (1993) have suggested, this failure to consider within-subject data may overlook some significant relationships between knowledge and interest variables that are obscured by group means. Employing within-subject techniques, Anderson-Inman and Horney (1993), for example, were able to identify three prototypic hypertext users based on their comprehension and interest. Indeed, in their discussions of results, some researchers acknowledged their reliance on group comparison procedures and the potential influence this reliance may have had on their reported outcomes. Lanza and Roselli (1991), for example, reported no significant differences between hypertext and more structured computer-based instruction groups on posttest measures. However, the researchers noted the extreme variability in the hypertext distribution which suggested a masked differential in the way subjects navigated through that system.

Third, in the computer-based or hypertext studies that we examined, there were few significant gains in knowledge reported, although there were some

significant effects for interest and attitudes (e.g., Schare, Dunn, Clark, Soled, & Gilman, 1991; van der Berg & Watt, 1991). We attribute this trend to the fact that, in these nonlinear text studies, subjects were not only operating in complex domains (e.g., genetics or statistics) but that they were also required to navigate through very complex computer-based systems. In other words, the compounding effect of a complex domain within a complex instructional system resulted in minimal effects on knowledge acquisition (e.g., McGrath, 1992). It is encouraging, however, that subjects' interest in the domain or their attitudes toward the computer-based system were often positively influenced.

Fourth, as others have (Schiefele et al., 1992; Tobias, in press), we noted the inconsistent findings for the relationship of knowledge and interest variables in the studies we reviewed. As we will discuss further in the section that follows, this inconsistency could be the result of any number of factors, particularly the varied and sometimes questionable way that these two constructs were measured. However, what surprised us even more, and what serves as the basis for this final observation, is the realization that so few studies specifically considered the interaction of knowledge and interest. While we were able to locate a number of investigations in which both variables were present, their interrelatedness was rarely embraced as an important or central issue by the researchers. Yet, it would seem that these interactions would hold the key to unraveling the apparently inconsistent picture of how knowledge and interest collectively enhance or inhibit human learning and development.

Basic Premises

In the previous section, we summarized and analyzed patterns in the identified literature. We now offer premises regarding the interplay of subject-matter knowledge, linear and nonlinear text, and interest that may guide future theory development, educational research, or instructional practice. Specifically, the bases from which these premises were derived are threefold. First, some of these premises were aligned with trends that emerged when we synthesized and charted each of the 66 studies (see Appendix B)—trends that demonstrated some interplay between two or more of the variables of concern in this study (i.e., subject-matter knowledge, text, or interest). For instance, by examining Appendix B, we saw a fairly consistent relationship, reflected in one of the premises which follow, emerge between the age and ability of students and the type of text processed.

Second, we were also guided in the formulation of these premises by critical questions or guided hypotheses that appeared in prior reviews (e.g., Alexander & Judy, 1988; Hidi, 1990; Pintrich & Schrauben, 1992). For example, previous reviews have suggested that individual and situational interest may relate differently to the acquisition of subject-matter knowledge (e.g., Schiefele, 1991). This issue was one which contributed to the subsequent premises by giving us cause to address these questions or test the hypotheses put forward. Finally, we were influenced in the development of these premises by inconsistent outcomes involving interactions among knowledge, text, and interest variables, such as the puzzling relationship between knowledge and interest (e.g., Tobias, in press). In this case, our goal was to see if our collection of interactive studies would shed new light on the question of whether knowledge and interest are positively and

significantly related. The information we found relative to this dilemma formed the basis for one of the premises we offer.

- When educators and educational researchers attend to the nature of domains and carefully evaluate learners' knowledge and interests accordingly, gains in subject-matter knowledge and interest are more apt to result.

One of the strongest advocates for considering the nature of domains in research and in practice is Spiro (e.g., Spiro & Jehng, 1990). Although none of his work with hypertext has been analyzed in this review for a number of reasons (e.g., not empirical), Spiro and his colleagues (e.g., Spiro, Feltovich, Jacobson, & Coulson, 1992) believe that hypermedia should be used to promote interdisciplinary, cognitive processing, especially in complex, ill-structured domains. Overall, in view of Spiro's argument that domains are inherently different, we were encouraged by the findings of several studies we reviewed (e.g., Anand & Ross, 1987; Carlsen & Andre, 1992; Hamrick & Harty, 1987; Long, Winograd, & Bridge, 1989; Romance & Vitale, 1992). In these studies, the researchers considered the nature of the domain, and students' understanding relative to that domain. They then attempted to orchestrate the learning environment to provide for maximum learning. Further, in these studies, the marked growth in students' subject-matter knowledge was accompanied by a concomitant rise in interest. For instance, not only did the fourth graders given in-depth, theoretically designed science instruction in the Romance and Vitale (1992) study learn more science than those following the school district's traditional curriculum, but their interest toward science and their self-confidence improved as well.

Similarly, Hamrick and Harty (1987) resequenced sixth graders' science textbooks so that the topics progressed from those of the physical sciences to those of the life sciences. Hamrick and Harty's purpose in restructuring the texts was so that the students could see distinctions between these composite domains. As a result of this restructuring, students' achievement and their interest in science improved more than they did for those whose texts were not reordered. While the work by researchers like Anand and Ross (1987), Romance and Vitale (1992), and others (e.g., Long et al., 1989) is commendable, a relatively small number of the researchers seemed sensitive to the nature of the domains from which they were drawing or considered this inherent nature to be important in their research.

Yet, when we consider the conditions of learning within schools, additional concerns about the nature of domains arise. Specifically, based on our findings, if we are to improve the state of learning for our school-aged populations, then we need to situate these developing minds in learning environments in which teachers are knowledgeable about the domain which they are instructing and sensitive to the domain-specific and general strategies required when students navigate within that environment. Teachers also need to be able to anticipate and assess students' misconceptions or faulty theories and be skilled at presenting important content in engaging and stimulating ways. As it currently stands, however, subject-matter instruction and its assessment continue to stress low-level, *inert knowledge* (Sherwood, Kinzer, Hasselbring, & Bransford, 1987; Whitehead, 1929), and classroom instruction is sometimes guided by individuals

who, themselves, possess limited understanding of the knowledge they are expected to convey to others (Alexander & Knight, 1993; diSessa, 1989).

There are some notable exceptions to this rather dismal observation. In a study of undergraduate students' use of nonlinear text in the domain of English literature (Beeman et al., 1987), for example, the professor was heavily involved in the interactions his students had with the learning environment. His goal was to promote complex thinking across multiple domains, rather than merely motivating individuals to use a sophisticated hypertext tool. As a result, students' hypertext compositions exceeded the professor's expectations. The training of more knowledgeable and committed teachers like those described in the studies by Beeman et al. (1987), Long et al. (1989), and Romance and Vitale (1992) is indeed a laudable goal.

- As individuals progress through the educational system, the impact of their domain-relevant knowledge to subsequent learning potentially becomes stronger.

As we looked across the studies in this review, it was apparent that researchers recognized that the depth and cohesiveness of individuals' subject-matter knowledge should increase as they move from elementary school, to high school, to college, and beyond. Evidence of this is the fact that the texts used in these investigations, both linear and nonlinear, became increasingly more identifiable as domain-related discourse, lengthier, and more specialized in topic. While elementary students read excerpts from literature or short passages about hobos (Hidi et al., 1982), click beetles (Garner et al., 1989) and inventors (Hidi & Baird, 1988), high-school and college students read chapters or extended passages about computer programming (Lanza & Roselli, 1991), genetics (Morrell, 1992), or quarks (Alexander, Kulikowich, & Schulze, in press-a).

Length of discourse was even more of an issue when the texts were nonlinear. Indeed, many of these computer-based texts were in excess of 250 pages (e.g., Egan et al., 1989; van der Berg & Watt, 1991). Further, many of the concepts studied within hypertext environments or assessed in the pretest or posttest measures represented specialized subfields within domains. For example, while the Gray (1989) and Gray, Barber, and Shasha (1991) studies fell within the domain of social studies, the particular choice of topics (i.e., poverty reduction, inflation control, and tax shelters) was more specifically associated with the subfields of sociology or economics. Likewise, the study by Guthrie and McPherson (1992), while representative of the domain of biology, focused exclusively on sport biomechanics. Moreover, the hypermedia tools used in several of the literature studies addressed such advanced topics as the interrelationship between literary and historical periods, such as the Romantic and Victorian periods in literature (e.g., Beeman et al., 1987).

What is also evident, particularly in the study by Tyler and Voss (1982), is that this subject-matter knowledge becomes a dominating factor when passage contents are high in information value. Therefore, when college students were asked to process demanding passages from *Pravda* that contained content incongruent with the students' expectations about USSR politics, it was the students with greater political science knowledge who garnered the most from the text. Interestingly, when the passages were lower in information value or contained more

congruent content, positive or negative views, more than knowledge, predominated. Tyler and Voss credited these results to the fact that the more knowledgeable students were able to process the incongruent passages at a greater depth than those students with less knowledge of the Soviet Union.

The outcome that Tyler and Voss describe coincides with what Stanovich (1986) describes as *Matthew effects*—aptly titled after the biblical text that reads: “For unto every one that hath shall be given, and he shall have abundance: but from him that hath not shall be taken away even that which he hath” (Matthew, XXV: 29). Thus, those with more subject-matter knowledge are better able to process or to navigate through text and are, therefore, able to gain substantially more domain-related information as they progress through school. However, those who lack such knowledge remain unable to tackle the increasingly more demanding exposition presented to them and continue to lose ground. To compound this situation, those with more domain knowledge continue to read and gain even more relevant subject-matter knowledge, while those without a satisfactory knowledge base fall farther and farther behind.

Coincidentally, it is interesting to note that, in those studies where knowledge was not found to be a significant factor in performance, the researchers typically controlled for prior knowledge (Schiefele, 1992b) or used materials which they assumed to be unfamiliar (Jegede et al., 1991). Thus, in these investigations, individuals were occupied in tasks for which they had little relevant subject-matter knowledge. It is not surprising that, under these circumstances, knowledge did not play a significant role in the learning outcomes.

- The use of linear or nonlinear text situates students differently within the learning environment and, thus, affects their interest and their ability to acquire domain knowledge.

One of the consistent observations supported by our review is that the decision to use linear or nonlinear texts contributed to learning environments that varied in significant ways. Those situational distinctions fall along several continua. We have labeled those continua as permanence to dynamism, individual to shared cognition, microprocessing to macroprocessing strategies, and intradomain to interdomain referencing.

In the studies we reviewed, when individuals read traditional text, they each encountered virtually the same written document (e.g., Hidi & Baird, 1988; Osaka & Anders, 1983). Further, even after processing, the textual artifact remained essentially unchanged. The words on the pages of Twain’s *Huckleberry Finn* first read by one of these authors in 1959 are the same as the words initially encountered by another author in 1975. In essence, there is a permanence to traditional, linear text. This does not mean that individual readers do not construct their unique interpretations of text; they, of course, do (Rosenblatt, 1983). Rather, the textual foundation of these interpretations in linear text remains quite consistent from one person to the next. The same cannot be said for nonlinear text.

In contrast to the more permanent character of linear texts, we reviewed a number of computer-based studies in which the texts were characteristically dynamic (e.g., Carver, Lehrer, Connell, & Erickson, 1992; McLellan, 1992). That is, the texts in these studies did not become an artifact, as such, until the

readers had personally interacted with them (Beeman et al., 1987). As students navigated through the hypertext environment, they combined texts in ways that suited their individual needs (McLellan, 1992). Additionally, they had the capacity to compose their own segments of text and add them to the textbase (Dillon, 1991; Turner & Dipinto, 1992). Thus, the *hyper-text* became an altered document, basically a new text, after each encounter. One's subject-matter knowledge and interest contributed to an unique, self-guided exploration of a wealth of stored information (Bolter, 1984, 1991b). This is what we mean by the dynamism of nonlinear text.

Not only are students able to manipulate and modify the textbase in hypertext studies in a way not typical for linear texts, but the transformations they make to text are directly and immediately available to students and teachers who are also situated in the learning environment. This shared experience during text processing helps to create what Purvis (1981) described as *shared significance* or what Pea (1989) characterized as *shared intelligence*. In other words, it is the nature of these hypertext systems to strongly encourage joint thinking and reflection about the text in order to foster shared cognition.

A somewhat different system exists in studies involving linear text processing (e.g., Sadoski, 1984). In these cases, students must often be prompted to share their thoughts and ideas when reading, because the tendency during linear text processing is for readers to operate in a more private, individualistic manner. Teacher-led or student-led discussions are one way that the more individual enterprise of processing linear text is made somewhat more public, although this interchange most often occurs after rather than during reading (Alvermann & Hayes, 1989). In summary, while linear texts tend to favor more individual cognition during the reading act, nonlinear texts seem to encourage more social cognition, at least in terms of current instructional usage.

Another distinction we noted between linear and nonlinear text environments is the type of strategic processing that was the apparent focus of students' efforts and researchers' inquiries. On the one hand, those reading or researching linear text seem to have a more microprocessing orientation, in that effort and interest are more apt to center on formulating inferences, distinguishing between important and nonimportant content, or on identifying main ideas at the paragraph or passage levels (e.g., Alexander, Kulikowich, & Schulze, in press-b; Garner et al., 1991; Jetton et al., 1992; Schiefele, 1992a; Schraw & Dennison, in press; Wade, Schraw, Buxton, & Hayes, 1993).

With nonlinear texts, on the other hand, emphasis is usually placed more at a global level of processing. The concern for subjects operating in these databased environments has typically been focused on their ability to navigate through the complex system (e.g., how they access and combine textual resources) rather than on what they may be thinking or what interests them at the word, sentence, or paragraph level (e.g., Beeman et al., 1987; Trumbull et al., 1992). This more global perspective among nonlinear text researchers may simply result from the fact that those working with computer-based and hypermedia systems are more often computer researchers, interested in the technological dimensions of the research. It might also be associated with the fact that databased text systems are relatively new or that those engaged in linear text research have been at the process of studying strategic behavior longer than those conducting nonlinear

text research. Thus, it is understandable that researchers using hypermedia would first investigate learner-system interactions at a macrolevel. Perhaps, the longer computer-based text systems are in use or under study, the more likely it is that educators and educational researchers will fine-tune their interests in strategic processing.

For all intents and purposes, learners situated in these nonlinear text environments are assumed to have the microprocessing strategies required to make sense of print. To us, such an assumption is problematic. When learners get lost in hyperspace (Conklin, 1987), it may not be due solely to their difficulties in building a conceptual map or schematic representation of the hypertext program, as has been argued (e.g., McLellan, 1992). Instead, it could be related to their failure to comprehend what they read at a sentence or paragraph level. If readers are unable to negotiate print at the microlevel, it seems unlikely that they can be successful when maneuvering at a macrolevel. Using older, more competent readers, as many studies did, is one way to reduce the concern for microprocessing abilities, but it cannot eliminate it.

The final issue we would like to raise about linear and nonlinear text situations relates to the distinction between intradomain and interdomain referencing. Just as nonlinear text systems strongly encourage students to share their cognitive processing via computer networking, they also directly promote interdomain linkages (Carver et al., 1992; Gray et al., 1991). Thus, the student reading a base text in American history may be prompted by the computer to consider relevant material from literature, science, or geography. This is what we mean by interdomain referencing. While this type of cross-domain referencing is also possible for the readers of more traditional text, it is not expressly prompted by the system itself, as in the case of hypertext systems. Therefore, the nonlinear text aids in interdomain learning while the linear text demands that readers devise their own connections between domains.

As we have attempted to illustrate, the decision to employ linear or nonlinear text alters the learning environment for those who operate within it. When text is more traditional, the artifact of the text is more permanent, and processing tends to be more private and focused at the microlevel, with interdomain referencing at a more implicit level. By contrast, when text is computer-based and more nontraditional, the text is more dynamic and socially constructed, strategies are more oriented toward macroprocessing, and interdomain referencing is more explicit. Collectively, these conditions situate learners within varied environments that have the power to significantly affect not only their interests but also their ability to acquire domain knowledge.

- Interest, particularly one's personal investment in the topic or domain, stimulates depth of processing in the content and, thus, enhances subject-matter learning.

One of the salient points that emerges from this review is that interest plays an important role in students' subject-matter learning. However, the nature of that role and its significance to learning outcomes are clearly associated with the type of interest that is activated. Going back to Hidi's (1990) seminal review on interest, we wish to at least theoretically disentangle transient, situation-specific interest from interest that comes about as a result of an individual's long-term

personal investment in a topic or domain. As noted, Hidi (1990) refers to the former as situational interest and the latter as individual interest. While both may serve a role in subject-matter learning, the latter is apt to have more consistent and positive effects on academic performance. Indeed, as several of the studies in this review demonstrated, situational interest can have negligible effects on performance (Sadoski et al., 1993) and can even interfere with the processing of domain-related text (e.g., Garner & Gillingham, 1992; Wade & Adams, 1990).

Concerning subjects' investment in a topic or domain, several of the investigations we examined provided evidence of positive effects. For example, after studying nursing majors and education majors solving statistics and probability word problems, Ross et al. (1986) concluded that educators should strive to make mathematical materials more interesting and meaningful to students by including topics that are familiar to them. Likewise, Dillon (1991) reported that nonlinear text readers who were already familiar and interested in the topic under study—human factors—had a better sense of the text structure of the pertinent domain and, as a consequence, learned more. Dillon also found that, because of topic familiarity, students could expend less time struggling with the navigation through hypertext and could “unfold sections of interest” (p. 924) that facilitated their comprehension. Further, Schiefele (1991) and Sacks (1989) found that subjects' interest in topics contributed to complex, deeper comprehension and to more in-depth recalls that included a greater production of inferences, more main ideas, and greater cohesion. In their study of children's reading, Anderson et al. (1991) also determined that, when children read stories of interest to them, their processing was at a deeper and less superficial level.

In another study using nonlinear text, Anand and Ross (1987) interviewed middle-school students to learn about their interests and hobbies. Based on this personal information, the researchers constructed mathematics story problems with personally relevant information. Students whose story problems were aligned with their preferences and interests outperformed students who worked less personally relevant problems. In addition, the students given the “interesting” problems were more favorably disposed to mathematical problem solving than their counterparts. As one student noted: “I was surprised because they were using my name in those situations. I could easily relate to the problem. I enjoyed the math program” (p. 77). In contrast to the research just described, however, studies involving nonlinear text often focused on students' interest in or enjoyment of the technology. Thus, we concur with the issue raised by Schick and Miller (1993, p. 12):

In reference to learner attitude and motivation in hypertext environments, further studies should be done in order to determine if the perception of computer/video based environments as entertainment rather than bonafide opportunities for learning may in fact prevent learners from realizing the full benefits of hypertext documents.

Finally, we would like to reinforce a point made by Hidi et al. (1982) that is related to this discussion of interest and subject-matter learning. In the earlier stages of learning, narrative texts are prevalent tools for learning and instruction. As students move through the educational system, they encounter more and more exposition. With narrative texts, it seems that the constructs of interest and

importance are closely linked (Hidi et al., 1982). That is, aspects of the story structure that capture attention are linked to the main ideas and, therefore, aid the reader in comprehension. This is not the case with exposition, however, where interest and importance tend to diverge. Thus, processing narration may require less effort and less domain knowledge than exposition because interest (particularly situational interest) and importance are often at odds in informational text. As Garner et al. (1991) determined, when students read passages about Stephen Hawking containing personally involving information about his life-threatening illness or seductive details about a wager he made with a colleague, this highly interesting information pulled students' attention away from the more central content. Thus, it would seem that in their efforts to make subject matter more personally relevant, especially when dealing with expository text, educators must be sensitive to the type of interest (i.e., situational vs. individual) that is invoked.

- The most powerful and positive learning outcomes occur in those contexts where students' knowledge and interests are well matched to the nature of the learning task.

One of our most pertinent insights from this review has to do with the impact that was evidenced when subjects were either well matched to their learning environment, on the basis of their knowledge and interests, or ill matched. Specifically, when there was a match between subjects' avocations or vocations and the subject matter of the text, there were significant and positive outcomes reported. The Tyler and Voss (1982) study, described earlier, furnishes convincing evidence for this premise in that the more knowledgeable political science students performed well even on demanding and incongruent passages taken from their field of interest. Further, our own recent study involving premed and graduate students' processing of demanding texts in immunology is another case in point (Alexander et al., 1993; Jetton, Alexander, & Kulikowich, 1993). Students in that study who had a base of subject-matter knowledge and who saw the content as relevant to their career aspirations were highly interested in what they read and had exceptionally high scores on both structured and unstructured recall measures. Clearly, certain students in that study were very well matched to the tasks they were given.

When there is a poor match, however, subjects do not have the requisite subject-matter knowledge to process information effectively or efficiently. When domain knowledge is low, students must rely on whatever relevant conceptual knowledge they can muster. In addition, when a mismatch exists, students are handicapped in their ability to employ domain-specific strategies and must, as a result, bank on general text-processing or problem-solving strategies to see them through the task. In Schiefele's (1992b) study, we see a case of this mismatch in that computer science majors were asked to process a chapter from a psychology textbook dealing with "Emotions." One of Schiefele's goals was to ensure that his subjects had limited knowledge about the subject matter and that there was a wide variance in interest in the topic. Thus, there were those students in the study with some interest in the topic but with no relevant knowledge and those with neither interest nor relevant background knowledge. The fact that Schiefele did not find any significant effect for knowledge on comprehension may well be

attributable to this mismatch phenomenon. Mismatches between learners and the learning environment need not be so dramatic in order to affect performance, however. Sometimes even subtle changes can influence the learning outcome. We saw this in the study by Murphy and Ross (1991) in which eighth graders processed mathematics story problems. When students read problems in which the protagonist was of a preferred gender, their interest and performance were noticeably higher than when the protagonist was not of the preferred gender.

Further, when students are ill matched to the learning task at hand, their interest (even for the tantalizing tidbits) may be negatively affected or may be of insufficient magnitude to compensate for their knowledge deficits. This seemed to be the case for one group of students in our recent study (Alexander et al., 1993). In examining learner profiles, we found one cluster of students who possessed very little domain knowledge, who showed very little interest of any kind in the texts, and who, as a result, learned very little. We referred to this collection as our “We don’t know and we don’t care” group. Similar knowledge, interest, and recall profiles were recently reported by Lawless and Kulikowich (1993) in a study of hypertext navigation.

Conversely, if students have some foundation of subject-matter knowledge and perceive a topic or domain as relevant, they may be able to overcome temporary setbacks to comprehension or learning. That is, they will likely find it easier to persist even when learning becomes difficult. This pattern was also seen among our student profiles when certain groups consistently performed well even when the texts were constructed to be highly technical and less engaging (Alexander et al., 1993). This issue of effectively matching students to learning tasks, therefore, would seem to have direct implications for instructional practice. That is, students are frequently asked to participate in learning tasks for which they are ill prepared and which they perceive as irrelevant to their personal goals or aspirations. Based on this review, the expectations for domain learning arising from such conditions are dismal, indeed.

- If one is to understand the interplay between knowledge and interest better, then greater care must be taken to assess what individuals know and care about prior to any intervention and that information used to design effective learning environments.

As we stated earlier, few of the studies we examined in this review took what we consider to be a comprehensive and systemic look at academic performance by considering knowledge, text, or interest in any interactive or multidimensional way. When knowledge was assessed, it was frequently measured only after an intervention had occurred. While the measurement of interest was somewhat better, there still remained studies where initial interest was assumed or where only one or two general items represented the interest variable (e.g., Jegede et al., 1991). Of course, our expectations for what educators and educational researchers should know about their students or subjects at the outset of any instructional intervention are high. That we acknowledge. Still, if we are to seek answers to questions about how knowledge and interest interact, then a careful assessment of what learners already know and what interests them is requisite.

Further, we must analyze these variables not only unidimensionally but also multidimensionally.

There have been a few studies that have explored subject-matter knowledge in a complex and interactive way. A case in point is Salomon et al.'s (1989) investigation of seventh graders' metacognitive abilities with respect to text processing. At the outset of their study, Salomon et al. gathered an array of information about the students, including their standardized reading comprehension scores and their grade point averages in language arts and mathematics. In addition, Salomon et al. did not expose students to any texts until those texts had been systematically evaluated for reading difficulty and interestingness by teachers, by experts in literature, and by students of similar age. Only texts that were of high interest were ultimately included in the study. Once students' subject-matter knowledge and the motivational aspects of texts were carefully assessed, the researchers examined whether a computerized Reading Partner—a kind of computerized tutor—would help students become more mindful and metacognitive during comprehension. Not only did students exposed to the Reading Partner expend greater effort and spend more time reading the texts, but their posttest metacognition, comprehension, and essay compositions were also much better than those of controls. This study and others (e.g., Wade et al., 1993) demonstrate that, when researchers take care in the assessment of students' prior knowledge, in the construction of tasks, and in the consideration of students' interests, much can be learned about the acquisition of subject-matter knowledge.

In our own program of research, we have also sought to study subject-matter knowledge in a comprehensive and systemic manner. In several recent studies (e.g., Alexander et al., 1993; Alexander, Kulikowich, & Schulze, in press-a, in press-b; Kulikowich, Alexander, & Jetton, 1993), we have begun to explore the interplay of knowledge, recall, and interest variables. In each of these studies, we assessed what students' knew about the domain prior to the studies. That assessment focused on students' passage-specific knowledge (i.e., topic knowledge measure) or their general knowledge of the field (i.e., domain knowledge measure). We also used passages that were evaluated by experts and piloted on students of similar age/ability. Further, we gauged subjects' interest in the texts they read at the passage, paragraph, and sentence levels and assessed comprehension by means of structured and unstructured recalls and student interviews. Finally, we examined data quantitatively and qualitatively and employed both within-subject and between-subject designs. Overall, these studies have shown us that the interaction between knowledge, recall, and interest is significant. For example, we recognized that domain knowledge was the best predictor of student interest and recall. In addition, we have determined that students with high domain knowledge consistently produced the highest interest ratings and did the best on the recall measures. Further, when students were asked about their processing, we found that more knowledgeable students were more capable than less knowledgeable students of distinguishing between what was important and what was interesting even when processing demanding exposition.

Concluding Remarks

In the previous section, we advanced premises about the interaction of subject-matter knowledge with text and interest variables—premises that we hope

can serve as guides for future research and practice. The results of this review should encourage teachers and researchers alike to explore the multifaceted and multidimensional nature of academic learning. We concur with Brown et al. (1989) and Resnick et al. (1991) that, as members of the educational community, we must all pay greater heed to the interplay between learner and environment. Our goal in this review has been to contribute in small measure to this understanding. We have promoted the idea that learners' interactions within rich and complex environments can result in meaningful subject-matter learning when concomitant factors are carefully weighed.

Yet, as educational researchers, we have only begun to unravel the complexity of domain learning as it occurs within real classrooms with real students who are absorbed in sometimes meaningful, sometimes demanding tasks. In this review, we have recognized how the instructional materials that are components of the learning environment color the nature of learners' processing and influence what they remember or find interesting. As a case in point, witness the sophisticated nature of computer-based systems, like hypertext, and their potential to transform students' interaction with subject-matter information. Still, much of that potential has yet to be realized, although we caught glimpses of it in several of the programs we reviewed (e.g., Ross et al., 1986; Salomon et al., 1989). In truth, whether they were built around linear or nonlinear texts, what the best of these innovative programs did was challenge students to confront their misunderstandings (e.g., Carlsen & Andre, 1992; Morrell, 1992), build directly on their interests and preferences (e.g., Hidi et al., 1982; Murphy & Ross, 1991; Tyler & Voss, 1982), or stimulate pluralistic or interdomain thinking (e.g., Alexander et al., 1993; Beeman et al., 1987).

The ability to design and implement such innovative programs, however, seems predicated on one's skill to uncover what learners know and care about. Certainly, to know more about learners, our tools of assessment must extend beyond the commonplace multiple-choice item or the often-administered Likert scale. Interview techniques, such as those used by Wade et al. (1993), are one alternative means of examining the roles of knowledge and interest in text learning. We must also locate or construct other measures that allow students to demonstrate the breadth and depth of their declarative and procedural knowledge (Alexander et al., 1993) and the valences of their interests and emotions (Schiefele, 1991). Further, as educational researchers, we must be more creative in our analysis of resulting data. Group comparison designs, while they should not be abandoned, should be augmented with analyses of individual differences where minor shifts or major changes in knowledge or interest can be recognized and tracked.

Finally, it is apparent to us that, whether the focus of instruction is linear or nonlinear text, the teacher remains the most important instructional resource available to learners. Teachers enrich interactions within the learning environment by providing students with relevant experiences and with the substantial scaffolding that permits them to construct meaningful interpretations and assimilate new understandings. Therefore, the more teachers know about their students and the domains and tasks under study, the more likely they are to establish a learning environment that benefits those students.

In essence, if we, as members of the educational community, are to forge a better understanding of how students acquire competence or expertise within a

domain or why they are intrigued or enticed by aspects of the learning experience, then we must open avenues of communication between all actors in the learning enterprise. Further, we must seek to understand the dynamic interplay between those actors and the subject matter they are striving to teach or to learn.

The mission to look at the dynamic interplay of subject-matter knowledge and interest in the processing of text may sound too challenging or too complex. That would not make the endeavor any less vital, however. It may be easier to control for situational and motivational factors present in the learning environment, but such an overly simplified or sterilized picture of learning obscures the reality that exists in subject-matter learning. Consider one renowned physicist's admonition to embrace complexity in the study of astronomical phenomena:

If you want to see a very faint star, you should look a little to the side because your eye is more sensitive to faint light that way—and as soon as you look right at the star it disappears. (Waldrop, 1992, p. 319)

Perhaps, in future studies of the learning phenomenon, we should not set our gaze solely or directly on the stars—the isolated constructs that reside in the instructional environment. Instead, we should look at the images and patterns that are produced when these constructs operate in synchronicity in the learning environment. To do otherwise is to fail to apprehend the true richness and complexity of subject-matter learning.

APPENDIX A

Journals examined in this review

<i>American Educational Research Journal</i>	<i>Journal of Educational Psychology</i>
<i>American Psychologist</i>	<i>Journal of Educational Research</i>
<i>Applied Psychology: An International Review</i>	<i>Journal of Educational Technology Systems</i>
<i>Artificial Intelligence</i>	<i>Journal of Experimental Psychology: General</i>
<i>Behavior, Research Methods, Instruments and Computers</i>	<i>Journal of Experimental Psychology: Learning, Memory, & Cognition</i>
<i>British Journal of Educational Psychology</i>	<i>Journal of Medical Education</i>
<i>British Journal of Educational Technology</i>	<i>Journal of Nursing Education</i>
<i>Cognition</i>	<i>Journal of Nutrition Education</i>
<i>Cognition and Instruction</i>	<i>Journal of Psycholinguistic Research</i>
<i>Cognitive Neuropsychology</i>	<i>Journal of Reading Behavior</i>
<i>Cognitive Psychology</i>	<i>Journal of Reading Education</i>
<i>Cognitive Science</i>	<i>Journal of Research on Computing in Education</i>
<i>College Science Teaching</i>	<i>Journal of Research in Mathematics Education</i>
<i>Communication Education</i>	<i>Journal of Research in Music Education</i>
<i>Developmental Psychology</i>	<i>Journal of Research in Reading</i>
<i>Developmental Review</i>	<i>Journal of Research in Science Teaching</i>
<i>Discourse Processes</i>	<i>Knowledge</i>
<i>Educational and Psychological Research</i>	<i>Language Learning</i>
<i>Educational Psychologist</i>	<i>Linguistics and Education</i>
<i>Educational Technology Research and Development</i>	<i>Machine-Mediated Learning</i>
<i>Educational Technology, Training, and Instruction</i>	<i>Mathematical Education</i>
<i>Instructional Science</i>	<i>Motivation and Emotion</i>
<i>International Journal of Man-Machine Studies</i>	<i>Poetics</i>
<i>International Journal of Research on Mathematics and Science Instruction</i>	<i>Reading and Writing Quarterly</i>
<i>Journal of Abnormal Child Psychology</i>	<i>Reading Research and Instruction</i>
<i>Journal of Applied Behavior Analysis</i>	<i>Reading Research Quarterly</i>
<i>Journal of Applied Developmental Psychology</i>	<i>Research in the Teaching of English</i>
<i>Journal of Computer-Based Instruction</i>	<i>School Science and Mathematics Science Education</i>
<i>Journal of Early Adolescence</i>	<i>SIGCUE Outlook</i>
<i>Journal of Educational Computing Research</i>	<i>Technical Communication Quarterly</i>
	<i>Technical Writing and Communication</i>
	<i>TESOL Quarterly</i>
	<i>The Modern Language Journal</i>
	<i>Written Communication</i>

APPENDIX B

Summary of reviewed studies

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Asher, Hymel, & Wigfield (1978)	Mixed	75 5th-grade students	Linear	25 cloze passages from Britannica Junior Encyclopedia	Pretest During	Heading achievement test used as a covariate Cloze Passages	Pretest Posttest	25 color slides Reading enjoyment scale	1. Children who received the high-interest material expressed significantly more enjoyment than those who received low-interest material. 2. Children comprehended more of high- than low-interest material.
Hidi, Baird, & Hildyard (1982)	Mixed-Literature, science, social studies	I. 30 adults II. 5th-and 7th-grade students	Linear	2 narrative - "Toad" and "Hobo" 2 mixed - "Divers" and "Easter Island" 2 exposition - "Orienting the Map" and "Electricity"	Posttest	2 written free recalls (immediate and 1-week delay)	Pretest	Salient aspects of text	1. Interesting and important ideas were the same in narratives. 2. In mixed tests, there was no relationship between interest and importance. 3. Few ideas were even considered interesting in expository texts. 1. Students in both grades recalled more essential than nonessential idea units <i>except</i> for mixed texts. 2. After a delay, 7th-grade students couldn't retrieve essential idea units the same way as when they correlated with interesting ideas.
Tyler & Voss (1982)	social studies	41 political science undergraduates	Linear	Passages about Soviet Union written by graduate students	Pretest Posttest	24 fill-in-the-blank questions Recall test Recognition test	Pretest	1. 6-point Likert scale 2. Adjective checklist	1. Students with high knowledge had favorable attitudes toward the topic. 2. Attitude related to aggressiveness in the congruent condition, but knowledge related to aggressiveness in the incongruent condition. 3. Knowledge was correlated with recall for the neutral countries.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Walton (1982)	Mixed	91 11th-grade students	Linear	Eight expository passages	Pretest Posttest	15 multiple-choice questions Written recall, 20 comprehension questions, and Prior knowledge rating scale	Posttest	Interest rating scale	4. Recognition was related to knowledge for the incongruent but not for the congruent condition. 1. Prior knowledge related to readers' ability to infer passage content and recall information. 2. Interest does not appear to be strongly related to text comprehension.
Osako & Anders (1983)	Mixed	184 9th-grade students	Linear	Four expository passages on rock & roll music, super-natural events, politics, and classical music	Pretest Posttest	Prior knowledge survey 10 multiple-choice questions as comprehension test	Pretest	7-point Likert scale	1. Topic interest was a marginal predictor of comprehension. 2. Students' comprehension scores were higher for the two passages rated as low interest.
Benware & Deci (1984)	Psychology	43 introductory psychology students	Linear	Titled "The Other Side of the Brain"	Posttest	1. 24 items of conceptual and rote learning 2. Questions	Pretest	Questionnaire	1. Students who studied text to teach others scored higher on interest and knowledge than those who studied the text to take a test. 2. Groups did not differ on rote learning.
Faircloth (1984)	Home economics	68 secondary home economics students	Linear (control group) CAI Nonlinear (treatment group)	Text related to consumer credit	Pretest Posttest	17-item knowledge test 17-item knowledge test (both immediate and delayed)	Unknown	Likert scale on interest in computers	1. Subjects who received computer-assisted instruction scored higher on the knowledge posttests than subjects who received traditional reading materials. 2. No relationship between interest in the computer and knowledge gained from CAI.
Sadoski (1984)	Literature	26 3rd-and 4th-grade students	Linear	Science fiction story by Arthur C. Clarke	Posttest	1. Retelling of story 2. Cloze passage	Posttest	Interest rating of 1 to 5	1. Knowledge of story structure predicts total recall. 2. Reader interest does not strongly predict knowledge of story structure. 3. Knowledge does not predict recall.

Baldwin, Peleg-Bruckner, & McClintock (1985)	Mixed	52 7th-and 8th-grade students	Linear	10 expository passages from the encyclopedia on various topics	Pretest Posttest	Multiple-choice test 10-item multiple-choice test	Pretest	Topic interest inventory	1. Students with high prior knowledge performed well on the knowledge posttest. 2. Students' scores on the knowledge posttest were better with high topic interest.
Steele (1985)	Science	127 11th-and 12th-grade students	Linear	2 expository science passages	Posttest	Written recalls	Unknown	Interest in the topic	1. Prior knowledge significantly affected recall. 2. Interest did not significantly affect recall.
Ross, McCormick, & Krisak (1986)	Statistics and probability	2 experiments 80 undergraduate nursing students. 50 undergraduate education majors	Linear	Story problems based on 4 introductory rules in statistical probability. Written in the following contexts: abstract, education, medicine, sports.	Posttest	1. Achievement test-20-item posttest, 5 items/rule. Also: transfer task 2. Recall test of specific content in problems	Pretest Posttest	Individual interest based on preferred context of story problem Attitude questionnaire 6-item Likert measure: "was interesting" and "was relevant to my interests"	1. For Exp. 1, 36/40 nursing majors preferred medical context in learner-choice group. Abstract least popular. 2. Treatments had little impact on subjects' recall of specific information. 3. For those in which material adapted to preference, significant main effects for "material was interesting" and "material relevant to my interests". 4. For education majors, sports and education most popular. 5. Significant main effects for recall of story problem, information, and thematic information. 6. No significant effects for attitudes.
Anand & Ross (1987)	Mathematics	96 5th-and 6th-grade students	Nonlinear Computer-assisted instruction	Story problems on fractions adapted from textbook, workbook, and teacher-made materials.	Pretest Posttest	2 items 11 open-ended mathematics items with 3 subsets: context, transfer, and recognition	Pretest Posttest	Biographical questionnaire Attitude scale: Likert format	1. Personalized group outperformed abstract group on context subtest. 2. Personalized group outperformed abstract group on formula recognition. 3. Attitudes were more positive for personalized-context group than concrete-context group.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Beeman, Anderson, Bader, Larkin, McClard, McQuillan, & Shields (1987)	English literature and plant biology	41 undergraduates in English and biology	Nonlinear Hypertext	<i>Intermedia</i> : A hypertext tool for promoting non-linear thought (exposition). 2 goals: connectivity of materials and visualization of concepts and ideas. Materials can be linked, annotated, and navigated.	Posttest	Observation done on classes, students were interviewed and asked to keep time and task diaries, grades recorded.	Pretest	Hypertext documents are aligned with students' avocational/vocational interests	- For English, 5 students received final grades on essays in excess of the maximum for the examination. These 5 students used <i>Intermedia</i> most often. - More As in biology as compared to year before: 44% vs. 34%. 100% of students felt that they learned "more" or "much more" than in an average course at their university.
Hamrick & Hartly (1987)	Science: physical science and life science	203 6th-grade students	Linear Resequenced vs. not resequenced	Resequenced text: expository textbook material resequenced by topics that would give students a better sense of the sciences: matter-life-climates of the world.	Posttest	Science achievement: Iowa Test of Basic Skills	Posttest	- Children's Attitude Toward Science Survey: 20-item Likert scale - Children's Interest in Science Measures 20-item Likert scale	- Significant differences in resequenced over non-resequenced for achievement, attitudes, and interest. - Increased interest in science among learners for whom resequencing was done may have a more refined insight as to what makes distinctions between various sciences/how they relate/ and what may make one science a better choice for a vocation over another science.
Bleakley, Waterberg, & Hopkins (1988)	Literature	540 5th-grade students	Linear	Adventure, mystery, and humor stories - adapted	Pretest Posttest	- Standardized reading test 8 four-option multiple-choice (mostly literal) - Cloze test, from story read	Posttest	8 questions of 2-4 options about the story	- Interest ratings were higher for girls. - Interest ratings were higher if main character was male. - Interest dropped when girls read story with male main character. - Interest dropped dramatically when males read story with female main character.

Hidi & Baird (1988)	Social studies	44 4th-grade students 66 6th-grade students	Linear	Famous inventors in history: expository, base, salient, and resolution texts	Posttest	2 written free recalls (immediately and 1-week delay)	Pretest	Salient and resolution texts	5. Mystery stories were higher than adventure or humor in interest. 6. No difference between interest and ability readers. 7. Girls performed better on comprehension measure. 1. Significant main effect for grade on recall. 2. Significant 3-way interaction: grade x text x time of recall. 3. When texts were modified, recall only improved for the more personally involving information, not the scientific information. 4. Students recalled more from salient text than base or resolution text.
Egan, Remde, Landauer, Lochbaum, & Gomez (1989)	Statistics	20 university students who had experience in statistics	Linear and non-linear Traditional text vs. hypertext	562-page text on data analysis and graphics	Posttest	1. Search performance 2. Open-book essays 3. Incidental learning: recall of headings	Posttest	Subjective ratings toward use of hypertext and interest toward science	1. Students using Super-Book text significantly answered more search questions correctly than students using linear text. 2. Essays of hypertext area judged to be better than essays of linear text users. 3. Superbook users were more interested in statistics following the experiment.
Garner, Gill-ingham, & White (1989) Study I	Science	20 graduate students	Linear	3 expository passages on insects	Posttest	1. Recall protocols of important information. 2. Classification of insects	Posttest	Rate passage on 5-point scale	1. Seductive details disrupt macroprocessing. 2. Little variability in interest ratings.
Study II	Science	36 7th-grade students	Linear	3 expository passages on insects	Posttest	1. Recall protocols of important information	Posttest	Assumed the text contained seductive details	1. Recall was poor. 2. Students who were given texts with seductive details recalled less important information than students who read texts with no seductive details.

APPENDIX B (*continued*)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Gray (1989)	Social studies	80 undergraduate students	Nonlinear Computer-based instruction with control over branching and no control	Students learned about policies related to poverty reduction and inflation control by making policy related decisions and then receiving textual and graphic feedback on the results of the decisions.	Posttest	1. 10 questions to find answers while searching through the screens. 2. Retention measure after screens turned off. 3. Number of times each of 10 policy decisions contained in the instructional material was examined.	Pretest Posttest attitudes	1. Locus of control Likert scale boring/interesting/enjoyable/not enjoyable	3. Redundant signaling increased recall. 1. Significant main effects for both locus of control and degree of learner control over branching on ability to locate answers to a list of questions. 2. Less control over branching = greater retention. 3. No significant effects on attitude toward computers-based instruction.
Long, Winograd, & Bridge (1989)	Science	26 5th-grade students	Linear	1 poem, 2 narratives, 2 expository passages on oceanography	Pretest	1. Reading achievement. 2. 15 multiple-choice questions.	Posttest	Rate passages on 5-point scale	1. Narrative passages rated highest for interest. 2. Comprehension, although low, was highest for narrative passages. 3. No correlation between achievement test and interest. 4. No correlation between comprehension and interest.
					Posttest	3. Multiple-choice comprehension test.			
Sacks (1989)	Unknown	55 7th-and 8th-grade parochial students	Linear	Unknown	Pretest	Prior knowledge pretest	Pretest Posttest	Interest inventory Rating scale	1. Subjects with high interest scored higher on the knowledge posttest. 2. Differences in subject's prior knowledge did not affect their comprehension scores. 3. Subjects with high interest generated more thoughts relating to the text topic than those with low interest.
					Posttest	Multiple-choice; cognitive processes questionnaire			
Salomon, Globerson, & Guterman (1989)	Literature, science, and current affairs	74 7th-grade students	Linear and non-linear Traditional text and computer	11 texts (150–350 words long) included for the study received	Pretest	1. Standardized test of reading comprehension.	Pretest	Texts were evaluated by experts as being interesting on a	1. Metacognition group took more time to read texts than did content questions and control

			erized reading tool		ratings by the teachers, experts in youth literature, and students for interest.	Posttest	1. Metacognition reconstruction task. 2. Reading comprehension test. 3. Essay writing task (also, math and lang. grades; reading time; self-reports of mental effort expended in reading)		3-point scale (0 = low to 3 = high). All texts received a mean rating of 2.0+ for interest	group. Reading Partner helped students be mindful. 2. Metacognition group had significantly more reported effort than other 2 groups. 3. Metacognition group scored higher on metacognitive reconstruction and comprehension. 4. Ratings of essays were significantly better for the metacognition group than content questions group and control group.
Pokay & Blumenfeld (1990)	Mathematics	283 geometry high-school students 9–12	Linear		Geometry proofs	Pretest During Posttest	Questionnaire on strategy use Achievement test grades	Pretest During Posttest	Motivation questionnaire	1. Early in the semester, value in algebra did not predict achievement. 2. Late in the semester, value in geometry predicted achievement on Test 1 but not on Test 2.
Wade & Adams (1990) Study I	Social studies	52 undergraduates	Linear		Passage about Horatio Nelson taken from encyclopedia	Pretest	1. Self-rating 7-point scale 2. Nelson Denny Reading Test	Pretest Posttest	Topic interest: 7-point rating scale Rate sentences on 4-point scale	1. Interest and importance are highly correlated in biography. 2. The higher reading ability groups' importance and interest ratings were higher.
Study II	Social studies	48 undergraduates	Linear			Pretest Posttest	1. Self-rating 7-point scale 2. Nelson Denny Reading Test 1. Free recall 2. Delayed recall	Pretest	Topic interest: 7-point rating scale	1. High reading ability group scored higher in recall. 2. Subjects remembered more unimportant than important information. 3. Low importance/high interest were recalled the most. 4. The two most memorable details were seductive details and main ideas. 5. Important factual information was poorly recalled. 6. Subjects remembered more interesting than uninteresting information.

APPENDIX B (*continued*)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Anderson, Wilk- inson, & Mason (1991)	Literature	149 3rd-grade students	Linear	4 stories adapted from children's magazines/ books	During Posttest	SRA test 1. Open-ended recall: letter to friend 2. Short-answer questions	Posttest	1–4 ranking	1. Children's interest rat- ings and recall were higher when they read for meaning versus sur- face emphasis. 2. Content and form were significant to story inter- est.
Cobb, Wood, Yackel, Nicholls, Wheatley, Tri- gatti, & Perl- witz (1991)	Mathematics	338 2nd-grade students	Linear	1. Frame activ- ities (not tex- tual). 2. Balance tasks. 3. Story prob- lems. 4. Symbolic sen- tences.	Posttest	1. Achievement test on con- cepts and ap- plication 2. Computation test developed by teachers	Posttest	Questionnaire	1. The students taught in the treatment group performed better on the knowledge measure. 2. No difference between the two groups on inter- est in mathematics.
Dillon (1991)	Human factors	I. 12 professional researchers II. 8 professional researchers	Linear and non- linear Traditional text and hypertext	2 articles selected from 1 journal in the field of relevance to the researchers	Posttest	1. Text assess- ment according to the struc- ture 2. Time 3. Error types	Pretest	Passages selected based on indi- viduals' voca- tional interest area.	1. Researchers had a clear sense of introduction/ method/results/discus- sion format. 2. Could assemble faster without headings than with headings. 3. Error types: confusion about introduction/dis- cussion distinction. 4. Subjects had little com- prehension of text's meaning. 5. Time with paper jour- nals faster than com- puter screen. 6. Exp. 2: 6/8 performed better or as well with electronic text suggest- ing a possible time/accu- racy trade-off.
Frau, Midoro, & Pedemonte (1991)	Science (geology)	36 high-school students	Nonlinear	Scientific papers, graphics, and digitized audio stored on a hard disk. Stu- dents chose be- tween a self- service mode and a task- guided naviga- tion mode.	Pretest Posttest	Concept map on concepts related to earthquakes 1. Interviews 2. Concept map	Pretest Posttest	Questionnaire Questionnaire	1. Students' choice of the first topic in a module depends on their inter- est. 2. Students were very stra- tegic in trying to figure out how system oper- ates. 3. Communication and ways of accessing infor- mation do not neces-

Garner, Alexander, Gillingham, Kulikowich, & Brown (1991) Study I	Science	48 undergraduate students	Linear	Stephen Hawking and his scientific work	Posttest	1. Free recall 2. Provide a title to the passage 3. Short-answer questions	Pretest	Assumed the text had seductive details	sarily guarantee an improvement in learning. 1. Ideas rated as high interest/low importance (seductive details) and as moderate interest/moderate importance were more frequently recalled than ideas rated as low interest/high importance. 2. Students who read generally more interesting text recalled more information than students who read uninteresting text.
Study II	Science	48 undergraduate students	Linear	Stephen Hawking and his scientific work	Posttest	1. Free recall 2. Provide a title 3. Short-answer questions	Pretest	Assumed the text had seductive details	1. More important generalizations were recalled for interesting text as opposed to uninteresting text. 2. Interest influenced recall. 3. High domain-knowledgeable students recalled more and answered the posttest knowledge questions more accurately.
Gray, Barber, & Shasha (1991)	Social studies	80 undergraduate students	Linear and nonlinear Traditional text and hypertext	Dynamic text on laws and tax shelters which enabled students to select topics of interest.	Pretest Posttest	1. Search task: find correct information 2. Search time 3. Grade point average	Pretest	1. Rotter's internal locus of control. 2. Assignment to dynamic text which is individual-interest-focused.	1. Dynamic text users found more correct information. 2. GPA correlated with number of correct answers. 3. Internal locus of control people were faster than external locus of control for both dynamic text and paper text.
Jegede, Okebukola, & Ajewole (1991)	Biology	64 Nigerian high-school students	Nonlinear CAI	Expository software presented texts with graphics followed by question and answer sessions and a point-scoring reward system.	Pretest	Knowledge and comprehension levels of Bloom's taxonomy 40-item multiple-choice test that focused on topics in computer lessons-	Pretest	20-item Likert scale "Using computers in a biology class will make biology more fun."	1. Students who used the computer in a group setting as opposed to individually or no computer had highest attitudes. 2. No significant effects for achievement.
					Posttest		Posttest		

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Lanza & Roselli (1991)	Computer science	60 undergraduate students	Nonlinear Hypertext, and CBI (computer-based instruction)	Text was based on 2 units, procedures and functions, needed in Pascal programming. Expository text included graphics.	Posttest	10 questions/5 per topic.	Posttest	Likert scale attitude scale measured: attractiveness, enjoyableness, and stimulation	1. No significant difference on learning measure. 2. However, more variability in hypertext groups suggesting greater individual variation. 3. 63% of control (CBI) students found CBI boring and unattractive. 4. 76% of hypertext found program stimulating and attractive, although 40% suffered from being disoriented.
Monroe (1991)	Literature	Middle school students	Linear	Stories about environmental problems	Posttest	Survey of comprehension	Posttest	Rating scale	1. Interest was a strong predictor of knowledge.
Murphy & Ross (1991)	Mathematics	252 8th-grade students from rural and suburban areas	Linear	Story problem sets built around preferred gender characters. Angie, the flight attendant, and Mack, the bird dog trainer.	Posttest	Students given 8 story problems to solve.	Pretest	1. Asked what kind of story set they would prefer to work on. 2. Attitude scale: Likert format	1. Significant protagonist type x school interaction: For the rural school, preferred protagonist group greater than the mixed protagonist group on problem solving. 2. 72% of students who received preferred gender problems indicated that they liked the materials compared with only 37% of those who received nonpreferred set and 59% who received mixed set.
Schare, Dunn, Clark, Soled, & Gilman (1991)	Medicine	83 undergraduate nursing students	Linear and non-linear CAI	Expository text was part of an interactive videodisc program called "Diabetic Health Assessment"	Pretest Posttest	Paper-and-pencil objective test Paper-and-pencil objective test	Pretest Posttest	Semantic differential scale Semantic differential scale	1. Students' achievement under interactive video was not significantly different than traditional lecture. 2. Students in CAI had more positive attitudes than students in traditional lecture.

Schiefele (1991) Study I	Psychology	53 male under- graduates	Linear	Titled "Emo- tions" (book chapter)	Pretest Posttest	Control for prior knowledge 12 open-ended questions	Pretest	Questionnaire	1. Interest influences com- plex and deep compre- hension questions. 2. Prior knowledge did not weaken effect of inter- est.
Study II	Social science	Senior high-school students	Linear	Titled "Prehistoric People" and "Television" out of encyclopedia	Pretest Posttest	Control for prior knowledge 12 open-ended questions	Pretest	Questionnaire	1. Low interest subjects had more verbatim rep- resentation. 2. Controlling for prior knowledge did not re- duce interest effect. 3. Prior knowledge had nonsignificant effects.
van der Berg & Watt (1991)	Statistics and hy- pothesis testing	88 juniors and se- niors in the communication sciences	Nonlinear Hypertext	125-frame, 285- page hypertext document on statistics and hypothesis test- ing.	Pretest Posttest	Classroom exam- ination Classroom exam- ination	Posttest	Questionnaire with fixed- and open- ended re- sponses	1. No consistent differ- ences between those who use hypertext sys- tem and those who do not. 2. Replacement group did not like learning with hypertext materials the way that the supplemen- tary group and control group did.
Carlsen & Andre (1992)	Science: elec- tronic circuits	83 undergraduate students. These students had not taken col- lege physics.	Linear and non- linear Linear text vs. conceptual change text	Linear text: adapted from middle-school/ high-school ex- pository texts. Conceptual change text: consisted of the linear text plus conceptual change sections that elicited stu- dents' precon- ceptions.	Posttest	26 multiple-choice items that asked conceptual questions about circuits.	Pretest	Questionnaire: 17 multiple-choice items about things like ma- jor, experience with electricity, and computers.	1. Experience/interest with circuits and computers had no impact on post- test scores. 2. Significant main effect for text type in favor of conceptual change text on knowledge measure.
Carver, Lehrer, Connell, & Erickson (1992)	American history	20 9th-grade students	Nonlinear Hypermedia	Text search and hyper author tool that con- sists of reflec- tion tools to help students structure their ideas about topics associ- ated with World War I or life- styles	During	Observation of students	Pretest Posttest	Likert scale	1. Students spend a great deal of time devoted to text/information design. 2. Significant increases in mental effort, interest, collaboration, planning, and individualization. 3. Finding information was highly correlated with consideration as to how to present information.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Gardner, Simmons, & Simpson (1992)	Science/meteorology	114 3rd-grade students	Nonlinear CAI	Expository <i>weather-school</i> interactive software package with text plus link with classroom activities seen as experiments.	Pretest Posttest	Knowledge of meteorology 13 questions: 6 declarative knowledge, 7 application knowledge	Pretest Posttest	Attitude scale: 15 items Likert-type test Included "attitudes toward science, computers, manipulatives"	4. Making graphics colorful was negatively correlated with planning, understanding relationships, and developing representation. 1. Hands-on activities group seemed to increase both knowledge of and attitudes of science. Computer-assisted instruction next followed by control group.
Garner & Gillingham (1992)	Science	36 undergraduates	Linear	Stephen Hawking and his scientific work	Pretest Posttest	Topic knowledge test 1. Recall 2. Questions: short-answer	During	Cognitive interest rating of each paragraph as high, moderate, or low interest	1. Pretest topic knowledge was low. 2. Subjects were moderately interested in the subtopics. 3. Subjects performed well on the posttest knowledge measures. 4. Seductive details did not influence performance on the posttest knowledge measures. 5. Topic knowledge and interest were positively related. 6. Interest and posttest knowledge were related. 7. Subjects who had moderate topic knowledge were more likely to be interested than uninterested in the text.
Guthrie & McPherson (1992)	Biology	45 undergraduates	Linear and nonlinear Traditional text and hypertext	Six units on sport biomechanics	Posttests	Format structure not mentioned	Posttests	Semantic differential and interviews	1. More than 80% of subjects gave a high positive response for how "encouraging" and "effective" hypertext was. 2. More than 80% of subjects gave a medium positive response for how "pleasant" and "interesting" hypertext was.

Jetton, Alexander, & White (1992)	Science	144 undergraduates	Linear	Stephen Hawking and his scientific work	Posttest	1. Free recall 2. Write questions about the passage 3. Delayed free recall	Pretest	Assumed the text had seductive details	3. Subjects who received hypertext with lectures scored higher on competency measures than subjects who did not receive hypertext, but not significantly. 1. Regardless of a given orientation, students recalled more seductive details and biographical information than scientific information. 2. Students remembered more biographical than scientific information regardless of given orientation. 3. Students were just as likely to write questions over paragraphs of high interest/low importance as paragraphs of low interest/high importance.
Kinzie, Sullivan, & Berdel (1992)	Science	164 9th-grade students	Nonlinear CAI	Computer-assisted instruction in two forms: learner control and program control. Expository text was adapted from "Energy Choices and Challenges"	Pretest Posttest	Standardized reading test 1. Time to complete 2. Multiple-choice test	Pretest Posttest	Students' preferences for science with and without computers Open-ended questions	1. Significant sex by control interaction showed that males performed better under program control than learner control. Females were better with learner control. 2. Subjects with higher reading ability took less time to go through text. 3. Preference for study of another subject went from 69%, when both options were available on computers, to 29%, when only science was available on computers.
McGrath (1992)	Mathematics	103 undergraduate students	Linear and nonlinear Hypertext/hypertext media as compared to paper text, menu-driven CAI, and NoMenu CAI	Hypertext had icon-based menu on every screen, designed for high user control	Pretest During Posttest	Survey of mathematics background and test for misconceptions in geometry Time on task Test for misconceptions in geometry and 3 surface area problems	Posttest	Likert scale	1. No significant differences found between groups on misconceptions test. 2. High spatial ability individuals in the paper text group took significantly more time to finish lesson than low spatial ability individuals in the hypertext group.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Morrell (1992)	Biology: photosynthesis and genetics	110 high-school students	Linear and non-linear Computer-assisted instruction vs. traditional lecture/discussion	Expository CAI included graphics plus text and was user-paced Content selected because it is often troublesome for students	Posttest	Multiple-choice test for photosynthesis unit=20 items Multiple-choice test for genetics unit=18 items plus Punnett Square	Posttest	Student attitude survey toward CAI and Likert items included: "CAI makes the subject more interesting" and "CAI makes the material less interesting"	3. No significant differences were found for enjoyment of lessons, although the low spatial-NoMenu group gave the most favorable rating while the high spatial-paper group gave the least favorable rating. 1. No significant difference in achievement between photosynthesis CAI and control group. 2. No significant difference in achievement between genetics CAI and control. 3. Significant difference between genetics CAI and photosynthesis. Traditional in favor of no CAI. Argue that genetics had many new terms and abstract concepts. 4. Photosynthesis group found CAI more motivating and interesting than classroom instruction with regular texts. 5. Genetics group did not like CAI, and they did not do well with it.
Morrison, Ross, & Baldwin (1992)	Mathematics	240 6th-grade students	Nonlinear Computer-based instruction	An expository instructional unit on the metric system	Posttest	1. Paper-and-pencil drawing items 2. 17 computer-presented items to measure knowledge and performance	Pretest Posttest	Pilot sample rankings of interest themes revealed that animals, sports, and clothing most preferred Attitude survey/ Likert scale	1. Sports items were most preferred followed by animals, clothing, and no context. 2. 90% of students strongly agreed that they liked being able to pick different topics for items. 3. No significant effects on posttest achievement found as a result of context selection.

Romance & Vitale (1992)	Science	128 4th-grade students	Linear	Course textbook tradebooks on science topics	Posttest	1. Iowa Test of Basic Skills in reading scores 2. MAT science test	Posttest	Likert scales in science and reading	1. The experiment performed better on ITBS and MAT. 2. The experiment displayed more positive attitudes toward learning science and greater confidence in learning science. 3. Experimental displayed more positive attitudes toward reading.
Scardamalia & Bereiter (1992)	Science	Study 1: 28 5th- and 6th-grade students	Linear	Reference materials on endangered species and a unit called "Wildlife in Jeopardy," the topic of endangered species, was considered familiar to students	Pretest	Standardized Achievement Test Generating questions about endangered species	Posttest	Raters asked how interested they would be in pursuing answers to students' questions	1. Knowledge-based questions received higher ratings of interest than text-based questions. 2. Knowledge-based questions received higher ratings for knowledge advance than did text-based questions.
					Pretest				
Schiefele (1992a)	Social studies	127 high-school seniors	Linear	Text 1 = prehistoric people Text 2 = television. Both adapted from World Book Encyclopedia	Pretest Posttest	1. SAT or ACT scores 2. Topic knowledge with multiple-choice Recognition test for each text	Pretest Posttest	Topic interest designed to measure emotional valence and personal meaning Quality of experience rating scales	1. Interest has a negative correlation with verbal ability. 2. Interest has a positive correlation with prior knowledge on prehistoric passage and television passage. 3. Interest significantly affects verbatim and propositional representation, but no effect on situational representation. 4. Only interest facilitates positive experience during reading. 5. Prior knowledge not at all significantly related to any component of text representation.
Schiefele (1992b) Study I	Psychology	53 computer science undergraduates	Linear	Titled "Emotions" (psychology book chapter)	Pretest	Prior knowledge (control variable) Comprehension- 6 simple, 3 complex, 3 deep comprehension	Pretest	Topic interest	1. Interest exerted a greater influence at deeper levels of comprehension. 2. No difference between high and low interest groups on prior knowledge.
					Posttest				

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Study II	Psychology	43 Social science undergraduates	Linear	Titled "Communi- cation" (psy- chology book chapter)	Pretest Posttest	Prior knowledge (control vari- able) Comprehension- recognition test of verbatim, propositional, situational rep- resentation	Pretest	Topic interest	3. Prior knowledge had no influence on compre- hension. 1. Subjects who displayed higher interest had fewer verbatim repre- sentations. 2. Low interest subjects were more likely to judge correct sentences false than high interest subjects. 3. Prior knowledge was significantly correlated with interest but had no influence on compre- hension. 4. The effect of prior knowledge was rela- tively independent of in- terest effect.
Schumm, Man- grum, II, Gor- don, & Doucette (1992)	Mixed	77 undergraduate students	Linear	10 intact passages from develop- mental reading text	Pretest Posttest	50-item multiple- choice test Self-written multi- ple-choice and essay questions over material read	Pretest	Self-ranking scale from 1–10	1. When topic interest and prior knowledge were high, subjects were more able to predict the questions professors would ask over the reading material. 2. Subjects with high topic interest were better able to focus ideas suggested by the structure of the text. 3. Subjects with high prior knowledge wrote clear, unambiguous questions over the reading mate- rial.
Signer (1992)	Social studies	78 4th-and 5th- grade students paired in dyads for an inter- group competi- tion	Nonlinear Computer-based instruction (CBI) with in- teractive video	Stories about events during the Revolution- ary War, <i>Begin in Boston</i>	During Posttests	During study: ob- servations, pa- ragraph comple- tion; post- tests; interviews; and teacher ques- tionnaires	Posttest	Exit interviews and question- naires	1. Students enjoyed typing in their own paragraph responses. 2. Students wrote that working with a partner made the program bet- ter, more fun, easier. 3. High reading ability stu- dents gave greater ap- proval for working in

Trumbull, Gay, & Mazur (1992)	Biology—specifically, cultural entomology	41 undergraduates	Nonlinear Hypermedia	“Bughouse” includes text, video, commercial movie clips, photographs, animation, and music about information associated with anthropology, art history, and entomology	Pretest Posttest	Grade point average Questionnaire related to use of search strategies and finding information in hypermedia	Posttest	Questionnaire about interest in topics students wished to explore further	pairs than low reading ability students. 12 students commented that “Bughouse” was interesting while another 10 were surprised with respect to the amount of material it contained. - Students of differing abilities used different search strategies. Those with the highest grade point averages used indexing strategies while those with lower grade point averages tended to be browsers.
Turner & Dipinto (1992)	Science	37 7th-grade students	Nonlinear Hypertext	Research reports about mammals	Qualitative study with ongoing evaluations	Observations, teacher interviews, student writings, and evaluations of student hypercard stacks	Ongoing evaluations	Students' reflections in writing	- Students expressed enthusiasm about sharing their ideas. - Hypercard input into hypertext helped students organize information about animals in a schematic way. - Students enjoyed writing information on stacks.
Wade (1992)	Social studies	Undergraduates	Linear	Revised passage of Horatio Nelson	Posttest	Attention measure; Recall measure	Pretest	Assumed the text had seductive details	- Seductive details were recalled more than main ideas or important factual ideas. - Readers spent twice as much time on important factual details as main ideas but remembered them the least. - Readers spent over 50% more time reading seductive details than main ideas.
Yang (1992)	Social studies	52 11th-grade students	Linear and nonlinear CBI-computer-based instruction and print-based instruction	CBI = 100 microcomputer screens Print-based text = 100 passages	Immediate and delayed posttest	16-item multiple-choice test for recall	Posttest	Instructional Materials Motivational scale: 30-item Likert scale that measured motivation and continuing motivation/interest	- On immediate posttest: CBI significant difference over PBI. 2-week later delayed posttest: no significant difference in recall. - CBI significantly more motivation and continuing motivation than PBI.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Alexander, Jetton, & Kulikowich (1993)	Physics and human immunology	33 undergraduate and graduate students	Linear	2 expository physics passages: S. Hawking and quarks adapted from popular magazines 2 expository immunology passages: bacteriophages and viral nucleic acids	Pretest Posttest	2 domain knowledge multiple-choice tests Unstructured and structured recalls	Posttest	Passage rating of interest	1. Cluster analysis of variables revealed 4 variable clusters: knowledge, interest, familiar recall, unfamiliar recall. 2. Cluster analysis of individuals revealed 6 different individual prototypes. 3. Relationships among knowledge, interest, and recall were similar for both domains.
Anderson-Inman & Horney (1993)	Language arts	31 8th-grade students	Nonlinear hypertext	"Electrotext Authoring System:" 44 pages of short stories; study focused on a story called <i>The Landlady</i> .	Pretest: vocabulary knowledge Posttest: free recall of story	Pretest paper and pencil Posttest open-ended format	Posttest	During study: the material selected, <i>The Landlady</i> , was considered interesting because of its surprise, suspense ending. Posttest: Attitude questions about use of hypertext	1. Three basic profiles of hypertext users emerged (i.e., "Book Lovers," "Studiers," and "Resource Junkies"). 2. Book lovers do not enjoy nonlinear text as they do linear text and do not recall a lot of information from hypertext. 3. Studiers used a lot of navigation strategies, enjoyed hypertext, and had high recall of story. 4. Resource Junkies were interested in the technology and not the story. They did not understand the story.
Barrett (1993)	Computer science—specifically, interface design	Senior undergraduates	Nonlinear Hypertext	"UID Tutorial" which describes interface design principles and shows good and bad examples of design	During course in computer science	Assignments and term projects	During-course in computer science	Informal evaluation of the tutorial	1. The interactive components of the hypertext were considered attention getters by the students to the point of distracting them from the static components of the hypertext. 2. Term projects demonstrated understanding and inclusion of tutorial ideas.

Christensen, Giamo, & Jones (1993)	Education: multimedia and instructional design	Undergraduate and graduate students	Nonlinear Hypertext	Hypertext as part of the "Drexel Multimedia Demo" shows Drexel 100 years ago as well as today	During the course of study Posttest	Questionnaires, interviews, videotapes Class projects and examinations	During development: design of the multimedia system was done to "capture people's interest" to present an attractive image of Drexel to the public	Questionnaires, videotapes, user commentary, interviews	1. 94% of users felt that the comprehensibility of the content was acceptable. 2. 39% of users felt that the material was difficult to access. 3. Student projects and class examinations indicated that they had acquired sufficient knowledge of hypermedia. 4. Students frequently requested access to system after initial use of the system.
Sadoski, Goetz, & Fritz (1993) Study 2	History	79 undergraduate students	Linear	Brief biographical paragraphs about historical figures	Posttest	7-point bipolar scale of content familiarity, immediate and delayed response	Posttest	7-point bipolar scale of interestingness	1. Concreteness and comprehensibility were highly related to recall, whereas familiarity was less so. 2. When a concrete sentence preceded an abstract sentence, the abstract sentence was recalled better than if it were preceded by another abstract sentence. 3. Concreteness was a significant predictor of recall, but interestingness, comprehensibility, and familiarity were not significant predictors.
Study 3	History	32 undergraduate students	Linear	Two 110-word and two 265-word paragraphs on Michelangelo and James Madison	Posttest	7-point bipolar scale of content familiarity	Posttest	7-point bipolar scale of interestingness	1. Concrete paragraphs were rated as more concrete, more interesting, and more comprehensible than abstract paragraphs. 2. Children comprehended more of high- than low-interest material.
Schraw & Dennison (in press) Study 1	Literature	60 undergraduate students	Linear	5-page narrative	Posttest	Free written recall task	Posttest	Ratings of interest obtained from undergraduate students	1. No difference in overall recall of the 3 groups. 2. Segments of text were more interesting than others, regardless of perspective.

APPENDIX B (continued)

Author (date)	Domain	Subjects (academic level)	Text type	Text description	Knowledge (pretest/posttest)	Knowledge how measured	Interest (pretest posttest)	Interest how measured	Outcomes
Study 2	Literature	46 undergraduate students	Linear	5-page narrative	Posttest	Free written recall task	Posttest	Ratings of interest prior to recall	3. Segments relevant to one's perspective were recalled more. 1. Interest ratings were affected by reader perspective. 2. Interest did not significantly affect recall. 3. Segments relevant to one's perspective were recalled more.
Wade, Schraw, Buxton, & Hayes (1993) Study 1	History	43 undergraduate students	Linear	Two 100-word texts on life of Horatio Nelson	Pretest	Nelson Denny Reading Test Self-rating 7-point scale	Pretest	Self-rating 7-point scale	1. Subjects recalled more unimportant than important information. 2. Subjects recalled more interesting than uninteresting information. 3. Highest recall was for high interest/low importance. 4. Lowest recall was for low interest/high importance. 5. Readers spent more time on important but uninteresting information and seductive details.
					Posttest	Free written recall task			
Study 2	History	30 undergraduate students	Linear		Posttest	Free written recall task Questionnaire	Posttest	Questionnaire	1. The same patterns of recall were obtained as in Study 1. 2. Readers use importance and difficulty to determine allocation of time and effort. 3. Readers spend little time on main ideas and boring trivia. 4. Readers' attention to seductive details is more impulsive and spontaneous.

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